

1st Year 2nd Semester Topology Exam

January, 2022

Work the following problems and show all work. Support all statements to the best of your ability. Work each problem on a separate sheet of paper. Each is worth 15 points.

1. Show that the torus T is not homeomorphic to the surface of genus 2, $M_2 = T \# T$.
2. Prove that the set of irrational numbers $\mathcal{I} \subset \mathbb{R}$ cannot be presented as a countable union of closed sets each of which has empty interior in $\mathcal{I}$.
3. Prove that $\pi_1(S^1) = \mathbb{Z}$.
4. Let CX denote the cone over the n -point space $X = \{1, \dots, n\}$. Show that every continuous map $f : CX \rightarrow CX$ has a fixed point.
5. Show that the fundamental group of $M = K \# T$ is not abelian where K is the Klein bottle.

Answer the following with complete definitions or statements or proofs. Each is worth 5 points.

6. Show that every map $f : S^2 \rightarrow \prod_{\alpha \in J} S^1$ of the 2-sphere to the product of circles is null-homotopic.
7. Show that for a retraction $r : X \rightarrow A$ the induced map $r_\#$ on fundamental groups is surjective.
8. Which of the following spaces taken with the product topology are locally compact? separable ?
 - (a) $\mathbb{N}^{\mathbb{N}}$;
 - (b) $I^{\mathbb{N}}$;
 - (c) $N^{\mathbb{I}}$;
 - (d) I^I .

Here $I = [0, 1]$ and $\mathbb{N}$ is the set of natural numbers.

9. State the Seifert-van Kampen Theorem. Can it be used to compute the fundamental group of S^1 ?

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10. Show that S^2 is not homeomorphic to S^4